

BUDHARAJU PAVANI SIVA PRIYA

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Professional Summary

Engineering undergraduate with strong foundations in Java, C++, SQL, object-oriented programming, and software development. Hands-on experience developing backend services, REST APIs, database-driven applications, and modular software systems. Experienced in software testing, debugging, API validation, and Linux-based development, with practical exposure to distributed software components and full-stack application development.

Education

Amrita Vishwa Vidyapeetham
B.Tech in Electronics and Computer Engineering

2027
CGPA: 9.01/10.0

Technical Skills

Programming Languages: Java, C++, SQL, C

Frameworks & Technologies: Spring Boot, REST APIs, MongoDB, MySQL, React, Node.js

Developer Tools: Linux (Ubuntu), Git, Postman, Docker

Core Concepts: OOP, DSA, Software Engineering, Distributed Systems, Operating Systems, Database Systems, Real-Time Systems

Experience

Cognizant
Software Engineering Intern

Jun 2026 – Aug 2026
Kochi, Kerala

- Developed modular backend services and full-stack web applications using Java, Spring Boot, and Thymeleaf, following structured software development practices.
- Implemented RESTful API controllers and service-layer business logic, integrating application components with MySQL for transactional data processing.
- Performed software verification using Postman by testing JSON payloads, invalid inputs, error handling, session management, and concurrent request scenarios.
- Debugged application-level issues by analyzing API behavior and validating interactions between backend services, database components, and client-side workflows.

Projects

AI-Powered Travel Recommendation System

- Developed a full-stack recommendation platform using MongoDB, Express.js, React.js, and Node.js with RESTful APIs for client-server communication.
- Implemented a K-Nearest Neighbors (KNN) recommendation engine to generate personalized travel recommendations based on user preferences.
- Validated API behavior using edge cases including malformed inputs and empty datasets, while optimizing MongoDB queries for responsive application performance.

Distributed Navigation Software System

- Developed modular C++ software using ROS on Linux, separating sensing, navigation, decision-making, and motion-control functionality into independently executing software nodes.
- Implemented message-based inter-process communication between software components to coordinate navigation decisions and system behavior.
- Designed software logic to process sensor inputs, maintain navigation state, handle dead-ends and boundary conditions, and generate reliable movement decisions.
- Analyzed ROS runtime logs and inter-node communication to troubleshoot failures, debug system-level issues, and improve reliability of interacting software components.

Achievements

- 1st Prize – International Conference on Robotics and Mechatronics (ICRM 2025):** Won first place in the robotics competition for designing and demonstrating an autonomous hexapod robot.
- GUJCOST (Robofest 4.0 & 5.0):** Qualified through Ideation and Proof-of-Concept stages and secured Rs.1,00,000 total funding across two consecutive years for autonomous robotics engineering projects.